Beta-delayed neutron studies of $^{137-138}$I and $^{144-145}$Cs performed with trapped ions

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Neutron emission following β decay influences the distribution of heavy elements produced in r-process nucleosynthesis, impacts the control and operation of nuclear reactors, and can serve as a sensitive probe of the nuclear-structure properties of neutron-rich nuclei. This article presents a detailed study of the β-delayed neutron emission properties of $^{137-138}$I and $^{144-145}$Cs performed by confining ions in the Beta-decay Paul Trap. The daughter ions following β decay emerge from the trapped-ion cloud with negligible scattering allowing reconstruction of the recoil-ion energy spectrum from the time-of-flight, and from which neutron-emission branching ratios and neutron-energy spectra can be deduced. The results for the ^{137}I and ^{144,145}Cs decays are consistent with previous results performed using direct neutron-detection techniques. In the case of ^{138}I, a branching ratio of 6.04(72)% is obtained, confirming the more recent branching-ratio results which are a factor of two larger than measurements made prior to 1978.

I. INTRODUCTION

For neutron-rich isotopes sufficiently far from stability, the β^--decay Q values become larger than the neutron separation energy, S_n, of the daughter nucleus. In these cases, the β decay may populate excited states above S_n which deexcite by neutron emission. This decay mode, known as β-delayed neutron (βn) emission, can become the dominant decay mode for isotopes sufficiently far from stability and therefore plays an important role in the understanding of nucleosynthesis and of the nuclear structure of neutron-rich nuclei. It is also important in applied nuclear physics [1] because it is energetically accessible for many fission products.

Roughly half of the isotopes of elements from iron to uranium are thought to be produced through the rapid neutron-capture process (r process) in which isotopes are made through repeated neutron-capture reactions and β decays [2, 3]. Beta-delayed neutron emission can significantly influence how the neutron-rich isotopes synthesized in r-process environments decay back to stability and provides an additional source of late-time neutrons. The resulting decay-chain shifts and subsequent late capture of these neutrons during freeze out can be substantial [4, 5] and require better understanding. To date, there have been few measurements of delayed-neutron properties for isotopes near the proposed r-process path and sensitive techniques are needed to reach these exotic, short-lived isotopes.

Reviews of βn properties [1, 6] highlight that a detailed understanding of reactor kinetics and reactor behavior under various accident and component-failure scenarios requires higher-quality nuclear data. Approximations suitable for light-water reactors (such as the assumption that the delayed-neutron and fission-neutron energy spectra are identical) are not acceptable for fast breeder reactors [7], necessitating more precise βn data [3]. Improved nuclear data would allow an accurately modeled delayed-neutron flux and energy spectrum for any fuel-cycle concept, actinide mix, or irradiation history.

High-quality βn measurements also provide constraints for nuclear-structure calculations [8] and phenomenological models [9–11] used to predict the decay properties for nuclei for which no data exist, as well as improve the accuracy and flexibility of the existing empirical descriptions of delayed neutrons such as the six-group representation [1]. Despite the importance of βn data, the existing nuclear data are limited, especially for the neutron-energy spectra, and some recent studies have revealed discrepancies in published results [12, 13].

The International Atomic Energy Agency (IAEA) has recently identified βn precursors in different mass regions that could serve as well-established "standards" for the purpose of data evaluation and measurement [14]. For

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the heavy-mass peak of the fission-product distribution, the IAEA selected the precursors ^{137}I and ^{138}I as standards. These isotopes are abundantly produced in the fission of actinides and therefore are important to nuclear-energy applications.

In the work presented here, the β decays of ^{137,138}I and ^{144,145}Cs ions were studied in detail using the Beta-decay Paul Trap [15], an open-geometry linear radiofrequency quadrupole (RFQ) ion trap designed for precision β-decay studies. The ions were held in vacuum, and the emitted β particles, γ rays, and low-energy recoiling nuclei were detected in coincidence. The βn branching ratios and energy spectra were inferred from the time-of-flight spectra of the recoil ions using the approaches first developed in Ref. [16].

The decay of ^{137}I had previously been studied using a variety of neutron-detection techniques and the most recent measurements have provided consistent results [17, 18]. The $^{144-145}$Cs decays had been studied using proton-recoil proportional counters [17, 19] and although the branching ratios had been determined to a fractional precision of 5%, the βn energy spectra had only been measured once.

For the decay of ^{138}I, however, the most recent measurements yielded branching ratios of 5.56(22)% [17] and 5.32(20)% [20], respectively, which are a factor of two larger than the measurements performed prior to 1978 [21, 22]. Additional results, especially from a measurement technique with a different set of systematic effects, were therefore desired to resolve this discrepancy.

II. MEASUREMENT APPROACH

The βn branching ratio and energy spectra of $^{137-138}$I and $^{144-145}$Cs were determined from the recoil energy imparted to the daughter ions following β decay. The BPT, was used to collect and suspend radioactive ions in vacuum at the center of a radiation-detector array. The detector array consisted of two ΔE-E plastic scintillator telescopes, two microchannel plate (MCP) detectors, and two high-purity germanium (HPGe) detectors, used to detect the β particles, recoiling daughter ions, and γ rays, respectively, emitted following the decay. Fig. 1 depicts the arrangement of the BPT and the detector array.

The recoil ions following β decay emerge from the trap volume essentially unperturbed by scattering allowing the identification and study of neutron emission from momentum-conservation considerations [16]. Decays for which a neutron has been emitted can be identified from the ions that have higher recoil energies than would be allowed from the emission of only leptons and γ rays. Because the momentum imparted by an emitted neutron dominates the nuclear recoil, the energy of the emitted neutron can be reconstructed using conservation of momentum. This novel way to perform delayed-neutron spectroscopy circumvents the challenges associated with direct neutron detection.

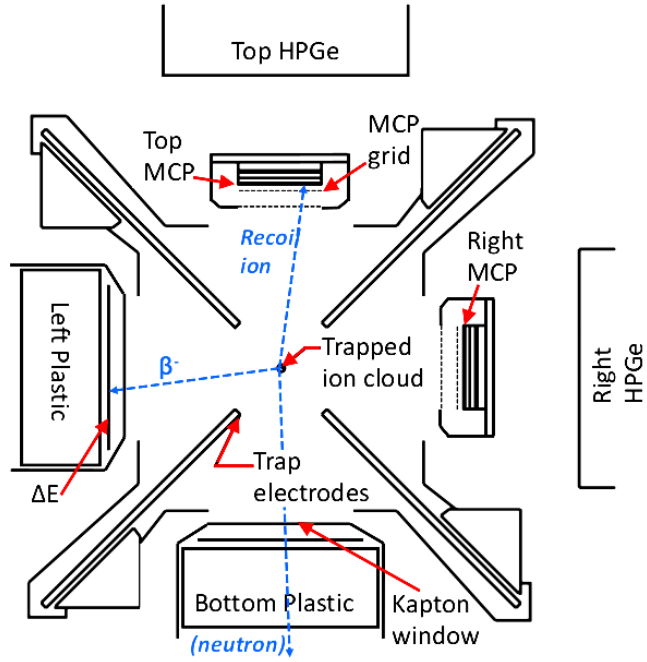

FIG. 1. (Color online) Schematic of the cross-sectional end view of the BPT experimental setup (not to scale). The dashed arrows represent an example of decay-product trajectories when a neutron is emitted.

The time-of-flight (TOF) of the recoiling ion is determined from the time difference between the detection of the β particle and the associated recoiling daughter ion using thin ΔE plastic-scintillator detectors and MCP detectors, respectively. In the TOF spectrum, as shown in Fig. 2, the higher-energy recoil ions (with kinetic energies up to tens of keV) characteristic of neutron emission have shorter TOFs than the other recoiling daughter ions (which have kinetic energies less than 500 eV). The narrow peak from nearly-simultaneous coincidences originates from the detection of β particles and γ rays and establishes the timing offset.

The βn branching ratios can be deduced by comparing the number of β-ion coincidences with these shorter TOFs characteristic of neutron emission to the total number of β decays from trapped ions obtained from the number of detected: (1) β particles emitted from the trapped precursor, (2) β-ion coincidences with longer TOFs characteristic of β-particle and γ-ray emission, and (3) β-delayed γ rays. The agreement obtained in the βn branching ratios when using these three approaches provides additional confidence that systematic effects associated with the measurement are sensible.

III. EXPERIMENT

The βn precursors $^{137-138}$I and $^{144-145}$Cs were produced at the CAlifornium Rare Isotope Breeder Upgrade (CARIBU) facility from the spontaneous fission of

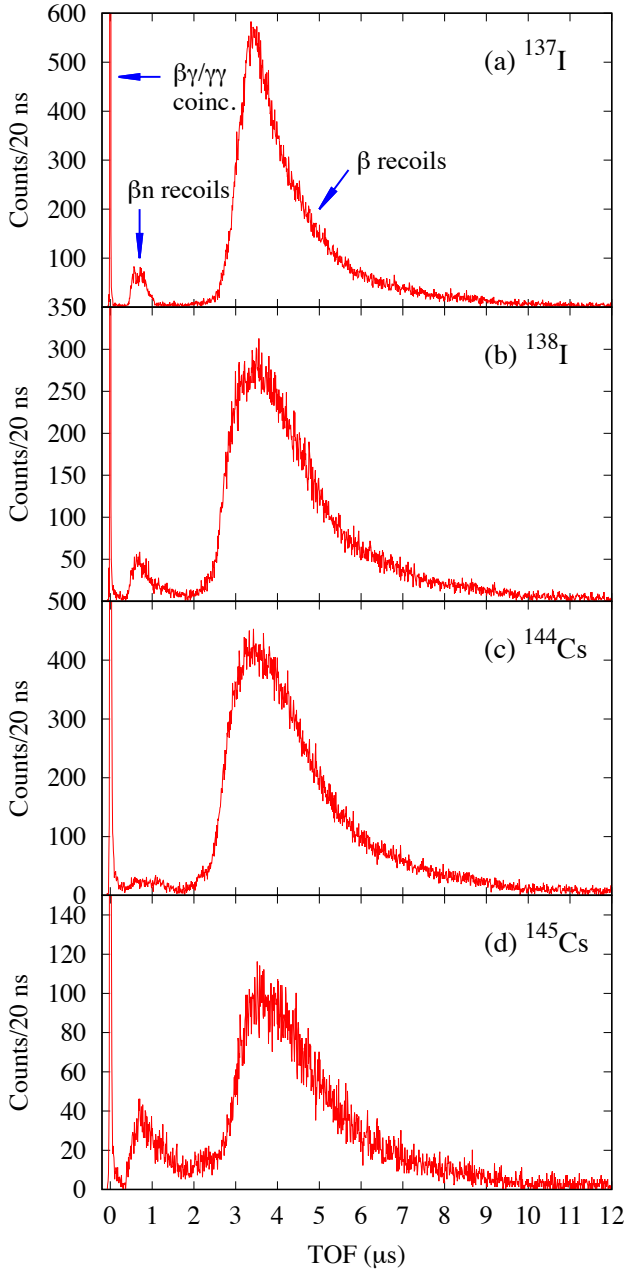

FIG. 2. (Color online) Recoil-ion TOF spectra with accidental background subtracted for all ΔE-MCP detector combinations. The coincidences between 0.3 and 2 μs are due to recoil ions having emitted a neutron, whereas the vast majority of the distribution above 2 μs is due to decays without neutron emission.

a ∼100-mCi ^{252}Cf source. The fission products emerging from the source are thermalized in helium buffer gas and extracted from the exit nozzle of the gas catcher [23] as singly-charged ions using a combination of gas flow and electric fields. These ions are then focused into the RFQ section where the residual gas is pumped away and the ions are cooled and subsequently accelerated to 35 keV for delivery to a high-resolution isobar separator [24]. The isobar separator was operated with a mass resolution of $M/\Delta M \approx 15000$ to select a single mass-to-charge ratio and suppress the isobars. In the cases studied, there remained only a small contribution from the nearest-mass isobars, which was accounted for in the analysis.

The selected ions were then accumulated in an ion guide and delivered to the BPT in bunches, where the ions were captured, cooled, and trapped. The BPT electrodes consisted of four sets of stainless-steel plates which come within 11 mm of the trap center and are segmented into three sections along the beam axis. The radiofrequency (RF) voltages confined the ions radially, while the direct current (DC) potentials provided the axial confinement. The applied RF voltage was measured directly using a high-voltage probe following the data collection. The amplitude was determined to ±1% precision and the time-dependence was found to deviate slightly from $V_{rf}\cos(2\pi f t)$, with f=310 kHz, and V_{rf} ≈100 V, and could be described by including higher harmonics at 620 kHz and 930 kHz contributing with amplitudes less than 10% of the magnitude of the primary frequency. The trap stability condition, determined with the Mathieu equations [25], was selected such that the β-decay daughter ions, which all have a charge state of 2^+ or greater, were not confined in the trap. The DC potential applied to the three sections along the beam axis was 20V, -17V, and 20V. To cool the ions in the ion cloud, helium buffer gas was injected into the center of the BPT. Analysis of the leading edge of the recoil-ion TOF distribution revealed that the ion-cloud spatial distribution was consistent with a Gaussian distribution of ≈1 mm [26] at full-width at half maximum (FWHM) when assumed to be identical in all three spatial dimensions.

Ions were accumulated in the BPT by maintaining a ∼5 V electrostatic valley at the center of the trap to retain the cold trapped-ion population while capturing a newly-delivered bunch. The accumulation periods ranged from 0.4 s for the ^{145}Cs measurement to as long as 6 s for the iodine isotopes. Following a set number of captures, the trapped ions were ejected from the BPT and collected on a silicon detector about 1 m further downstream to monitor activity levels. The signal observed in the BPT detector array during the period after the ion-cloud ejection was used to assess backgrounds that originated outside the ion cloud. The number and length of capture cycles and the length of the background measurements were chosen based on the radioactive halflife of the isotope being studied and the level of isobaric contamination. The average beam intensities for ^{137}I, ^{138}I, ^{144}Cs, and ^{145}Cs were about 800, 200, 600, and 100 ions/s, respectively.

Isobaric contaminants were identified and accounted for by means of the information from the β-γ coincidence spectra (using the γ-ray lines listed in Table II), as well as the time dependencies of the β-particle count rate buildup and decay in the ΔE-E telescope. The

beam content was also monitored periodically by performing a mass analysis with the Canadian Penning Trap Mass Spectrometer (CPT-MS) described in detail in Refs. [27, 28]. The CPT-MS data provided additional information on the relative amounts of isobars in the beam to confirm the decay-counting measurements.

In the ^{137}I dataset, the absence of the 455-keV γ ray from the decay of ^{137}Xe in the β-γ coincidence spectra indicated this isotope contributed less than 0.01% of the total trapped-ion decays. This is consistent with other observations during the experimental campaign of the gas catchers inefficiency at extracting Xe isotopes due to their high ionization potential. The ^{137}Te γ ray at 243 keV was observed in the β-γ coincidence spectra; however, because the γ-ray intensity for this transition is not known, the contribution from this isotope had to be determined from rate of buildup and decay of the β-detector signal. From this information, it was determined that ^{137}Te accounted for 20% of the total trapped-ion decays, a value which is consistent with the relative fission-product yields of ^{137}Te and ^{137}I from the spontaneous fission of ^{252}Cf. Similarly, for the data collected for ^{138}I, the absence of the γ-ray lines from the decay of ^{138}Xe indicated that this isotope contributed less than 1% of the trapped-ion decays. The decay of ^{138}Te has no known γ rays, but its yield is only 5% of the ^{138}I and the analysis of the buildup and decay of the β-singles signal limited the ^{138}Te contribution to less than 10% of the total number of trapped-ion decays. There was no evidence of isobaric contamination in the $^{144-145}$Cs beams. In these cases, the independent yield of the ^{144}Xe and ^{145}Xe isobars were only 2% and 0.1% that of the ^{144}Cs and ^{145}Cs isotopes, respectively. This was consistent with the β-singles analysis constraining the activity of these isotopes to less than 1–2% of total decays. The absence of characteristic γ-ray lines from the decays of ^{144,145}Ba in the β-γ coincidence spectra constrained the ^{144}Ba activity to less than 0.1% of the ^{144}Cs activity, and the ^{145}Ba activity to less than 1% of the ^{145}Cs activity.

Two resistive-anode Chevron-configured MCP detectors, each with a nominal active area of 50.3×50.3 mm^2, were used to detect recoil ions with 1-ns timing resolution. The MCP detectors, denoted as Right and Top to indicate their relative positions looking down the beam axis, were located 53 mm from the trap center. The front face of the MCP detector was biased to approximately -2.5 kV to accelerate the ions and improve the uniformity of the detection efficiency. A grounded, 89%-transmission grid located 4.5 mm from the front face of the MCP helped shield the detector from the RF fields and prevented the MCP bias voltage from affecting the recoil-ion trajectories until the ions were directly in front of the detector.

The relative amounts of charge collected at the four corners of the resistive anode allowed the determination of the ion impact position with sub-mm precision. The position calibration was established prior to data collection by observing the pattern that resulted from placing a mask on the surface of the MCP and illuminating it with a ^{238}Pu α-particle source. The entire assembly was mounted on a ceramic plate, and covered with a grounded aluminum case with a square aperture covered by a second 89%-transmission grid. The MCP detector housings were specially designed to compactly fit between the electrodes of the BPT and to allow HPGe detectors to be brought within 10 cm of the trapped ion cloud.

The intrinsic detection efficiency of MCP detectors is nearly independent of energy for keV-energy heavy ions [29]. However, the electronic threshold can affect the fraction of recorded events, giving rise to some energy-dependence to the detector response [29, 30]. Analysis of the pulse-height-distributions (PHDs) from recoil-ion impacts revealed that the Top MCP had a significantly lower gain than the Right MCP, and therefore was about 30% less efficient at detecting low-energy recoil ions than the Right MCP. The energy-dependent MCP efficiency correction is described in Sec. IV D. The intrinsic efficiency was determined from a detailed analysis of the ^{134}Sb β-ion coincidences [26] as mentioned in Sec. V A.

The β spectroscopy was performed with two ΔE-E detector telescopes (referred to as Left and Bottom, as shown in Fig. 1) made from EJ-204 plastic scintillator. Each detector consisted of a 1-mm-thick, 10.6-cm diameter ΔE detector located in front of a 10.2-cm-thick, 13.3-cm-diameter E detector capable of stopping the highest energy β particles from the fission-product decays. The ΔE detectors were located about 105 mm from the center of the BPT. The β particles were identified by energy deposition in the ΔE detector, as this thin detector has only a $\sim$1% intrinsic detection efficiency for γ rays and neutrons.

The light from the ΔE detector was piped to two 1.5-inch diameter photomultiplier tubes (PMTs) using light-guide strips wrapped in thin specular reflectors. The E scintillator was coated in a layer of diffuse reflector paint and attached directly to a 5-inch diameter PMT. To prevent outgassing into the trap vacuum, each detector telescope was supported in its own vacuum chamber, separated from the ultrahigh-vacuum environment of the ion trap by a 10-μm-thick aluminized kapton window. The vacuum in the detector region was kept below 10^{-3} torr. Energy deposition measurements from the conversion electrons from the ^{134}Sb isomer decay and from a spectroscopy-grade ^{207}Bi conversion-electron source indicated that the ΔE threshold for β particles was 76$\pm$24 keV and 62$\pm$30 keV for the Left and Bottom detectors, respectively.

Two coaxial single-crystal p-type HPGe detectors were mounted on the vacuum chamber in re-entrant ports. The Top detector (as labeled in Fig. 1) was a 140%-relative-efficiency crystal, while the Right detector was a 80%-relative-efficiency crystal. The photopeak efficiencies were determined using ^{60}Co, ^{133}Ba, ^{137}Cs, and ^{152}Eu γ-ray sources with activities calibrated to 1.5–2.5% and a ^{56}Co source to extend the calibration up to 3.3 MeV. The sources were positioned within 1 mm of the geomet-

ric center of the trap using a specially-designed support mounted onto the RF electrodes.

Signal from any of the detectors triggered the data acquisition system by opening a 22-μs coincidence window for recording the amplitude and timing of the signals from each detector as well as the RF phase of the trap voltage. The deadtime per event was determined to be 142 μs.

Detailed Monte-Carlo simulations were used to interpret the results, calculate detection efficiencies, and determine systematic effects. The recoil-ion trajectories were calculated using SimIon [31]. Geant4 [32] simulation package was used to determine the energy deposition in the plastic-scintillator detectors from β particles, γ rays, neutrons, and conversion electrons. The β-decay kinematics were simulated using an event generator originally developed to study β-ν angular correlations [33, 34] and later adapted for βn emission [16] and complicated β-decay schemes [26, 35]. The code assumes an allowed β-decay energy spectrum and incorporates the known β-decay transitions to excited states in the daughter nucleus (using data from NNDC [36] or from a simple model fit to the β-energy and recoil-ion TOF spectrum [35]) to generate the decays. The simulation takes into account the charge-state distribution of the recoil ions and the size of the ion cloud. These properties are determined from the analysis of the β-ion coincidence data as described in Ref. [26].

For the isotopes studied here, many of the transitions are first-forbidden and can potentially have any value of $a_{\beta\nu}$ between -1 to 1, and is often not well defined. For βn emission, the β-ν angular-correlation coefficient $a_{\beta\nu}$ was assumed to be that of an allowed Gamow-Teller transition ($-1/3$). For all other transitions, an approximation was made that all β-decay branches for a given isotope have the same (initially unknown) value of $a_{\beta\nu}$.

IV. NEUTRON ENERGY SPECTRA

The neutron energy E_n was deduced from the TOF of the recoil ion, t_{TOF}, and the hit position on the MCP surface which establishes the distance d traveled by the ion. The distance is approximated by a straight path from the trap center to the MCP grid, after which the ion acceleration into the MCP surface is calculated analytically. The value for t_{TOF} is also adjusted to account for the ion acceleration between the grid and the MCP surface. The E_n can then be determined from the simple relation

$$E_n = \frac{M_r E_r}{m_n} = \frac{\frac{1}{2} M_r^2 (d/t_{TOF})^2}{m_n} \qquad (1)$$

where M_r and m_n are the masses of the recoil ion and the neutron, respectively, and E_r is the recoil-ion energy. Ultimately, an additional small correction for the lepton contribution to E_n is also included for certain detector pairs as described in Sec. IV B.

Dataset	Background counts	Correction (%)
^{137}I	212(22)	6.8(7)
^{138}I	156(22)	7.2(10)
^{138}I	48(11)	5.6(13)
^{144}Cs	748(41)	31.1(17)
^{145}Cs	344(33)	11.2(11)

TABLE I. Number of counts attributed to coincidences arising from a background at short TOF, for the region above what is equivalent to 100 keV in neutron energy spectrum, summed over all the detector combinations. The correction is also listed relative to the total number of βn counts. The ^{138}I data was collected under two experimental conditions with varying capture and background periods, hence two values are listed.

The neutron-energy calibration is dependent on the distance between the center of the ion cloud and the MCP detector. This distance was determined to be 53.0(5) mm for both MCP detectors from a detailed analysis of the recoil-ion TOF spectrum following the β decay of ^{134}Sb [26]. These results were consistent with the distances of 52.5(3) mm and 52.9(3) mm determined from physical measurements of the placement of the detectors and electrodes. The value obtained from the recoil-ion TOF spectrum is used in the analysis and results in a 2% uncertainty in the energy determination. The neutron energy resolution varies with energy from 30-50 keV at the lowest neutron energies to 100-150 keV FWHM at 1 MeV (depending on the isotope). The resolution is largely determined by the 1-mm FWHM spatial extent of the ion cloud and the recoil imparted by the leptons.

The neutron-energy spectrum determined from the β-ion coincidences requires accounting for a number of backgrounds, efficiency factors, as well as energy shifts which arise from the recoil imparted by the leptons. The following subsections examine these effects in detail.

A. Backgrounds

The TOF distribution of the accidental coincidences was determined from the ΔE-MCP detector coincidences between 15 and 20 μs, a region devoid of true β-ion coincidences as indicated by both data and simulations.

The TOF spectrum recorded during the time period that the trap is purposefully held empty revealed an excess of ΔE-MCP coincidences at TOFs less than 1 μs. This background, also observed during the time the ions were present in the trap, was most pronounced between 50–200 ns, decreased with increasing TOF, and overlapped with the TOF region where the recoil ions from βn emission were observed. This background was likely due to the decay of activity that accumulates on the BPT or detector surfaces and was most evident in the ^{144}Cs TOF spectrum due to its long data collection time and

high beam intensity.

The number of counts and shape of this background was characterized by using the data collected without trapped ions. The background was then scaled to match the amplitude of this effect in the TOF region of the trapped ions absent true β-ion coincidences (TOFs shorter than ≈300ns). The resulting corrections for the isotopes under study varied between 6 and 31%, and are summarized in Table I.

B. Lepton-recoil energy shift

The neutron energy determined using Eq. 1 neglects the momentum imparted to the nucleus from the lepton emission. For β-ion coincidences detected with the ΔE-MCP detector pair separated by 180°, the β-particle momentum is roughly in the same direction as the neutron momentum and therefore contributes to the energy of the nuclear recoil. Hence, the neutron energy inferred using Eq. 1, which attributes the entire recoil energy to the neutron, results in an overestimation of the neutron energy. For the coincidences in the ΔE-MCP detector pair separated by 90° the β momentum is roughly orthogonal to the neutron momentum, and its impact on the recoil energy is negligible.

The size of the correction that had to be applied to the reconstructed neutron-energy spectrum for the 180° detector pair was estimated with simulations. By comparing the energies of the emitted neutrons generated following β decay to the energies reconstructed from the recoil-ion TOF spectrum, the energy shift was ∼15% for 100-keV neutrons and decreased to below 2% for neutrons with energies above 1000 keV.

C. Neutron-ion coincidences

The momentum of the neutron is nearly in the opposite direction of the recoil-ion momentum so the neutron likely passes through the ΔE detector opposite the MCP detector where the recoil-ion event was detected. The resulting neutron-ion coincidences can yield an additional ∼10% ΔE-MCP coincidences for the detector pair separated by 180°, despite the small intrinsic detection efficiency for neutrons. The response of plastic scintillators to neutrons, which is mediated through proton recoil, typically induces a roughly 5 times smaller light yield than β particles depositing the same energy [37]. Since these neutron-ion coincidences could not be distinguished from β-ion coincidences, they increased the total coincidence detection efficiency for the 180° detector pair.

The neutron detection threshold for the ΔE detectors was estimated from the measured ratio of ΔE-ΔE-MCP triple coincidences (which are expected to be predominantly from β-neutron-ion coincidences) to the ΔE-MCP coincidences. The neutron detection threshold in the simulation was adjusted until it best matched the experimental results for the isotopes studied here. The resulting neutron-detection threshold of 370(70) keV was consistent with the factor of 5 less scintillation light expected for neutrons compared to β particles.

D. MCP detector efficiency

The MCP detector efficiency has an ion-impact-energy dependence because the higher-energy recoils give rise to larger signals which are more likely to exceed the electronic threshold. Due to the Poisson nature of the electron-multiplication process in MCP detectors [38], the pulse-height distribution (PHD) from ion impact events can be described by a Gaussian distribution [39] with the centroid and standard deviation both dependent on the detector gain and the impact energy of the ion [29]. The centroid of the PHD was observed to increase linearly with the impact energy, which is consistent with the Parilis-Kishinevskii relation [40], while the width of the PHD was observed to be independent of the ion-impact energy. Following an approach similar to Ref. [39], the decrease in efficiency for both detectors was estimated based on the fraction of the PHD which falls below the threshold.

The detector gain, and therefore detection efficiency, was found to vary over the fiducial area of the MCP. To account for the spatial dependence, the MCP was subdivided into 4×4 square regions, to compromise between the spatial resolution and the available statistics. The PHD from each region was then separately fit to determine the parameters for the Gaussian distribution. Figure 3 shows the measured PHD and resulting model fit over the entire MCP detector area for the lowest-energy ions (the ions resulting from decays without neutron emission) which have the lowest centroid and therefore show the largest fraction of events below the threshold.

The fraction of counts above the discriminator threshold was calculated analytically from the established relationship between the ion-impact velocity and the PHD gain and the results are shown in Fig. 4. The analysis of this MCP-efficiency correction is described in detail in Ref. [41].

E. Coincidence detection efficiency

The likelihood of both a β particle and a recoil ion following β decay striking ΔE and MCP detectors, respectively, and generating pulses large enough to rise above the detector thresholds depends on the decay kinematics and therefore varies with the neutron energy. The scattering of β particles off the electrodes and other material within the chamber increases the β-particle detection efficiency by up to 30% above what would be expected from the detector solid angle alone. For β particles the detection efficiency is affected by the energy loss in the thin kapton window in front of the plastic detector and the

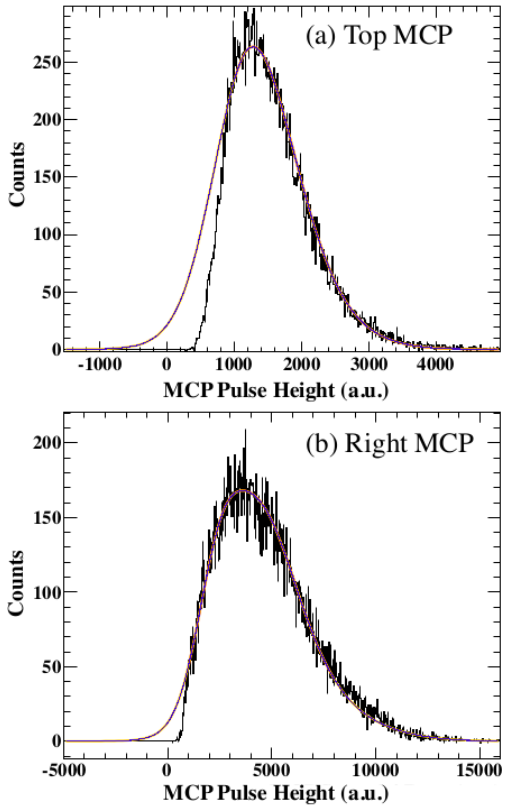

FIG. 3. (Color online) The PHD for ^{137}Xe ions following the β decay of ^{137}I striking the (a) Top and (b) Right MCP detectors with impact energies of 5 keV. The fit is a sum of Gaussian distributions which take into account the observed spatial variation in gain across the face of the MCP detector. This gain variation imparts a visible skew to the PHD. The spectra for the other isotopes are similar.

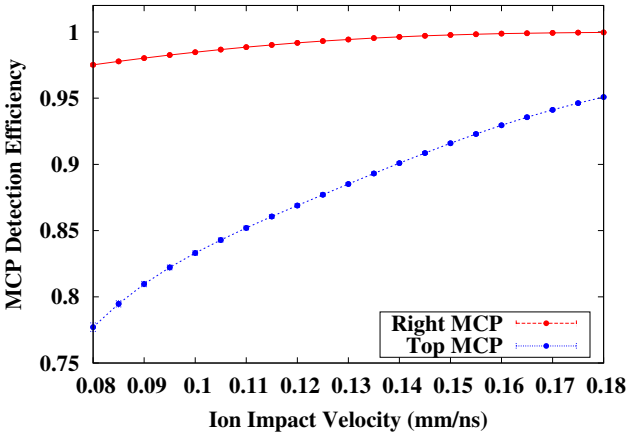

FIG. 4. (Color online) The calculated fraction of MCP detector pulses above the discriminator threshold as a function of ion-impact velocity. The recoil ions from β decays to the ground state or accompanied by γ-ray emission have impact velocities around 0.08 mm/ns, whereas recoil ions following βn emission have a range of energies that extend well above 0.1 mm/ns.

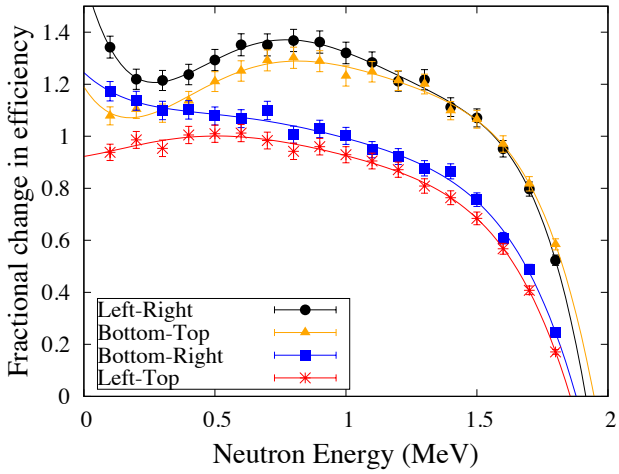

FIG. 5. (Color online) The coincident β-ion detection efficiency for the βn decay of ^{137}I for each ΔE-MCP detector pair, after normalizing to the product of the corresponding detector solid angles. The 180° detector combinations have higher efficiencies primarily due to the additional contribution from neutrons triggering the ΔE detector in coincidence with the recoiling ion. For the highest neutron energies, little energy remains for the leptons, and therefore the fraction of β particles detected drops rapidly; although this effect is large for decays populating excited states within 500 keV of $Q_\beta - S_n$, a negligible fraction of the neutron spectrum is typically found there due to the highly suppressed phase space for these decays. The lower efficiency of the Top MCP detector is also evident.

ΔE detector threshold. The ion trajectories, and therefore fraction of ions striking an MCP detector, is affected by the RF and DC fields. The MCP energy-dependent efficiency correction and the neutron-recoil ion coincidence correction, discussed earlier, also influence the total coincidence detection efficiency. Figure 5 shows the total β-ion coincidence detection efficiency for each ΔE-MCP detector pair for the case of ^{137}I decay.

The resulting neutron-energy spectra for ^{137,138}I and ^{144,145}Cs determined from the recoil-ion TOF spectra are shown in Fig. 6 and are compared to previous results that were directly measured using ^{3}He ionization chambers (^{137}I [42] and ^{138}I [43]), and proportional counters (^{138}I [44] and $^{144-145}$Cs [19]). In each case, the results confirm the existing measurements in the energy range above 100 keV. Below 100 keV, the βn recoils cannot be distinguished from the other recoil ions and therefore no information about the βn energy spectrum was obtained. The results obtained with the BPT are subject to different systematic effects than encountered using conventional direct neutron detection techniques. Direct measurements are typically affected by large backgrounds from scattered neutrons and γ rays, and require complex spectral unfolding due to the non-Gaussian detector response. These challenges are avoided in the recoil-ion technique, since the detector response is nearly Gaussian

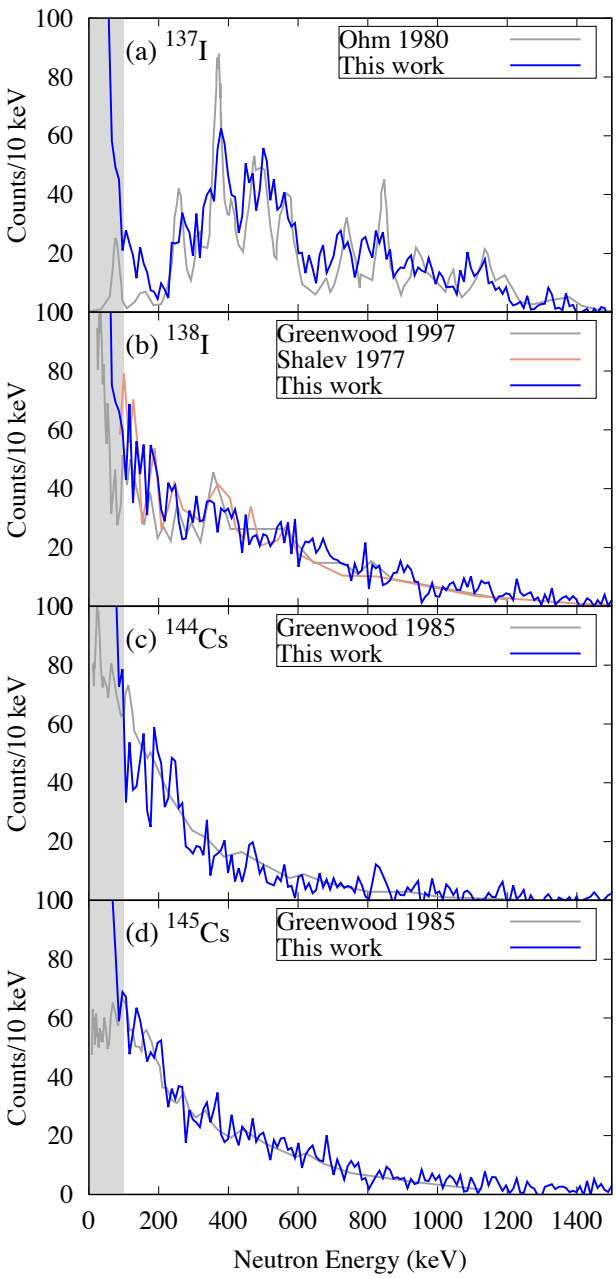

FIG. 6. (Color online) Neutron-energy spectra determined from β-ion coincidences compared to previous results from direct neutron-detection methods [19, 43, 44]. In the region shaded in gray, the βn decays cannot be resolved from decays without neutron emission.

and few backgrounds appear at the TOFs corresponding to the recoiling ions.

V. BETA-DELAYED NEUTRON BRANCHING RATIOS

For each isotope, the βn branching ratio, P_n, is determined from the ratio of the number of β decays resulting in neutron emission $N_{\beta n}$ to the total number of β decays N_β. The quantities for $N_{\beta n}$ were determined from the detected number of β-ion coincidences with TOF characteristic of βn emission $n_{\beta R}$ via:

$$N_{\beta R} = \frac{n_{\beta R}}{\epsilon_{\beta R} f}, \tag{2}$$

where the $\epsilon_{\beta R}$ is the total β-ion coincident detection efficiency and f is the fraction of the neutron energy spectrum above 100 keV. The values for N_β were determined with three different ways, through (1) the number of detected β particles originating from the trapped species of interest, n_β, (2) the number of detected β-ion coincidences with longer TOFs characteristic of lepton and γ-ray emission, $n_{\beta r}$, or (3) the number of detected β-γ-ray coincidences, $n_{\beta\gamma}$, after accounting for the efficiency of detecting these various decay signatures. In the following equations, R and r denote recoils from β decay with and without neutron emission, respectively. The results obtained with each ΔE-MCP coincidence pair was analyzed separately because of their different detection efficiencies.

A. P_n obtained from β-particle detection

Using the number of detected β particles, the P_n for each isotope can be determined from

$$P_n = \frac{n_{\beta R}}{n_\beta} \cdot \frac{\epsilon_\beta}{\epsilon_{\beta R}} \cdot \frac{1}{f}, \tag{3}$$

where ϵ_β is the β-particle detection efficiency, $\epsilon_{\beta R}$ is the detection efficiency for the β-ion coincidences characteristic of neutron emission, and f is the fraction of the neutron-energy spectrum above a threshold of 100 keV, for which the ion recoil energies could be unambiguously identified from neutron emission.

To determine n_β, the fraction of total ΔE triggers originating from the trapped species of interest had to be isolated from backgrounds from any neighboring isobars and any activity located outside of the ion cloud. This was accomplished by comparing the data to a model that takes into account the buildup and decay of the different species over the course of the measurement cycle while enforcing the decay-feeding relationships between different populations [45]. The time dependence of the β-particle signal rate over the course of the measurement cycle was sufficient to determine n_β to better than 5% precision in all cases.

The values of ϵ_β were determined for each isotope from simulations that take into account the β-decay transitions and subsequent deexcitation cascades, the geometry of the ion trap and detector array, and the detector

thresholds [36]. The $\epsilon_{\beta R}$ values for each ΔE-MCP detector pair were obtained from the efficiency curves used in interpreting the neutron-energy spectrum, like the ones shown in Fig. 5.

The intrinsic efficiency of the MCP influences $\epsilon_{\beta R}$ and was determined from a detailed study of the decay of trapped ^{134}Sb ions [26]. ^{134}Sb decays predominantly to the ground state of ^{134}Te via a first-forbidden 0^- to 0^+ transition, for which $a_{\beta\nu} \approx 1$ [46]. By comparing the results from the simulation to the data, the MCP intrinsic efficiency (which includes the ion loss in passing through the two 89%-transmission grids) was determined to be 29.5(13)% for the Top MCP, and 33.7(15)% for the Right MCP. The ratio $\epsilon_\beta/\epsilon_{\beta R}$ was determined to a precision of 7%.

The values for f were sourced from previous direct measurements of the neutron-energy spectra [19, 42–44] and were 0.96(1), 0.81(5), 0.69(4), and 0.76(3) for ^{137}I, ^{138}I, ^{144}Cs, and ^{145}Cs, respectively.

Table III lists the resulting P_n values attained with the β-particle analysis, each a weighted average of P_n results from the four ΔE-MCP detector pairs.

B. P_n obtained from the recoil-ion TOF spectrum

The P_n for each isotope can also be determined solely from the recoil-ion TOF spectrum, since the total number of decays is the sum of the neutron and β recoils. In this case,

$$P_n = \frac{\frac{n_{\beta R}}{\epsilon_{\beta R}\cdot f}}{\frac{n_{\beta R}}{\epsilon_{\beta R}\cdot f} + \frac{n_{\beta r}}{\epsilon_{\beta r}}} = \frac{n_{\beta R}}{n_{\beta R} + n_{\beta r}\frac{\epsilon_{\beta R}\cdot f}{\epsilon_{\beta r}}} \tag{4}$$

where $\epsilon_{\beta r}$ is the β-ion coincident detection efficiency for recoil ions without neutron emission.

The advantage of determining P_n solely from the recoil-ion TOF spectrum is that the intrinsic efficiency of the MCP detector cancels out in the ratio $\epsilon_{\beta R}/\epsilon_{\beta r}$. However, determining $\epsilon_{\beta r}$ is challenging because the recoil ions have energies less than 500 eV and are significantly perturbed by the electric fields of the BPT. Hence, $\epsilon_{\beta r}$ is sensitive to the details of the decay scheme, as well as the charge-state distribution of the recoil ions following the β decay.

To accurately determine the $\epsilon_{\beta r}$ values despite incomplete knowledge of the decay scheme details, an approach was developed as described in Ref. [35]. The method fits the simulation results to several β-ion coincidence properties measured during the experiment (such as β-particle energy and recoil-ion TOF spectra) in order to determine the fraction of ions which reach the microchannel plate detector to within ±4%.

The recoil-ion charge-state distribution was estimated from the variation of the β-ion coincidence rate as a function of the phase of the RF voltage of the ion trap. Appropriate weighting of the charge states is deduced by comparing data to the simulations, in a process described in detail in [26].

Isotope	γ energy (keV)	I_γ(%)
^{137}I	1218.0, 1220.0	16.3(16)
^{137}Te	243.4	–
^{138}I	588.8	56.0(54)
^{138}Xe	258.1	31.5(13)
^{144}Cs	198.9	46.5(25)
^{144}Ba	387.9	13.4(13)
^{145}Cs	175.2	19.8(24)
^{145}Ba	544.3	9.9(12)

TABLE II. The γ-ray transitions used to characterize isobaric contaminants through the analysis of β-γ coincidence data, with intensities from Refs. [36, 47]. For the ^{137}I transitions, the two γ-ray lines were not resolved, so the sum was used. No γ-ray transitions are known for the decay of ^{144}Xe and ^{145}Xe, and the absolute intensity of the ^{137}Te γ ray is not known.

The decay scheme information relevant to $\epsilon_{\beta r}$ includes the β-ν correlation coefficient $a_{\beta\nu}$, β intensities to specific decay branches, and the γ-ray decay scheme. However, this information may be either incomplete or unavailable. Consequently, an approach was developed, described in detail in Ref. [35], which can produce $\epsilon_{\beta r}$ values using only the data collected in the experiment. Specifically, the simulation parameters are adjusted until the ratio of recoils detected in coincidence with β's at 180° vs. 90° matches the observed one. The simulated β energy spectrum is also adjusted to match the E detectors spectrum. As demonstrated in [35] and [26], this method yields an $\epsilon_{\beta r}$ value precise to within 4% that is then used in the P_n calculation.

The P_n values obtained from the recoil-ion analysis are listed in Table III.

C. β-γ coincidences

The number of β decays can also be determined from the detected β-γ coincidences emitted by the trapped ions of interest. Therefore,

$$P_n = \frac{n_{\beta R}}{n_{\beta\gamma}} \cdot \frac{\epsilon_{\beta\gamma}}{\epsilon_{\beta R}\cdot I_\gamma \cot f}, \tag{5}$$

where I_γ is the absolute intensity of the γ-ray transition and $\epsilon_{\beta\gamma}$ is the coincident detection efficiency for the β particles and associated γ rays of interest.

The β-γ coincident events are free from isobaric contaminants, as the γ-ray transitions are unique for each isotope. The γ rays and I_γ values used to determine P_n

P_n method	Isotope ^{137}I	^{138}I	^{144}Cs	^{145}Cs
β	6.07(73)	5.77(75)	2.46(32)	13.03(160)
βr	7.18(83)	6.18(50)	2.95(24)	13.53(90)
$\beta\gamma$	7.09(120)	6.28(102)	2.29(41)	16.68(310)

TABLE III. Summary of the P_n results obtained with the BPT.

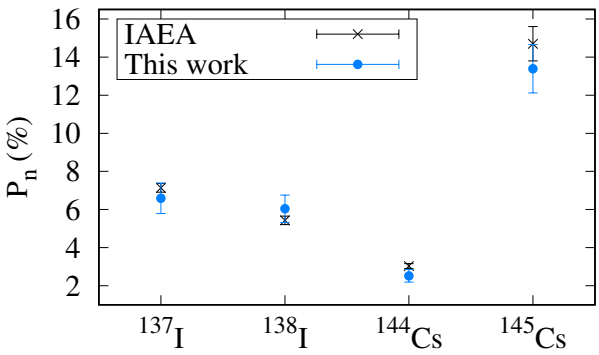

FIG. 7. (Color online) Comparison of the IAEA evaluation values and the weighted average of the three β-recoil methods.

in this work are listed in Table II. For ^{144}Cs, values from an IAEA technical report were used [47].

For each HPGe detector, ε_γ is less than 1% so the statistical precision of the measurement was limited. In addition, decays of the isotope of interest occurring outside of the ion cloud contribute a β-γ coincident background. This background was estimated from the data collected during the measurement cycle with the trap purposefully held empty. In the case of ^{137}I, the feeding from the decay of ^{137}Te also had to be taken into account. In addition, the uncertainties on I_γ, when available, are frequently 10% or greater. As a result, the P_n results obtained from the measurement of β-γ coincidences typically have the lowest precision of the three methods discussed here. Table III summarizes the β-γ analysis results.

D. Discussion of ^{138}I P_n results

The P_n results for ^{138}I are of particular interest as the most recent measurements [17, 20] indicated the P_n value is a factor of two larger than the results obtained from earlier measurements [21, 22]. The P_n results from this work are consistent with the recent measurements and indicate the earlier values can be excluded from the world average, therefore establishing ^{138}I as a reliable standard. The values in Refs. [21, 22] were determined by separately measuring β particles and neutrons from ^{238}U and ^{232}Th fission products, a method that relies on accurate assessment of the total β activity and hence is

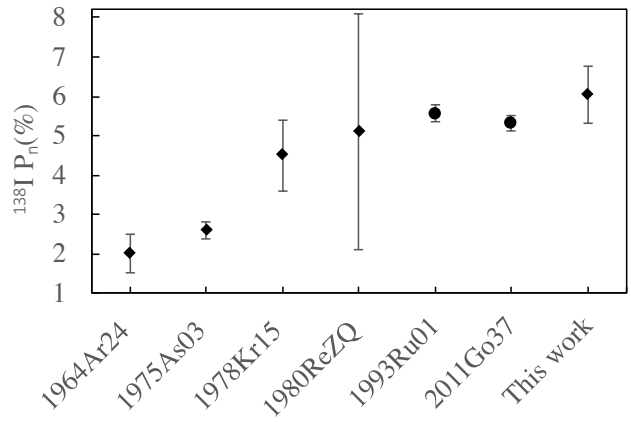

FIG. 8. Comparison of ^{138}I P_n values from [17, 18, 20–22, 48], with the results from this work.

subject to uncertainty from improperly-subtracted contaminants. Furthermore, the older data was normalized to now-obsolete fission yields. The comparison of the P_n values from various works is shown in Fig. 8.

VI. CONCLUSIONS AND OUTLOOK

The βn branching ratios and energy spectra for $^{137-138}$I and $^{144-145}$Cs βn precursors were determined by studying the momentum distribution of the recoiling ions following β decay. The measurements presented here for ^{137,138}I and ^{144,145}Cs were carried out by delivering mass-separated beams from the CARIBU facility to the BPT where the activity was collected and held in vacuum. Following β decay within the trapped ion cloud, the emitted β particles, γ rays, and low-energy recoiling ions emerged with minimal scattering and βn emission branching ratios and energy distributions were inferred from the recoil imparted to the nuclei. The measurements performed with trapped ions allow three methods to determine P_n by detecting the β particles, recoiling ions, and γ rays in coincidence. Each of these approaches is subject to different systematic uncertainties and yields consistent results with an uncertainty of about 10-20%. The method of obtaining P_n directly from the ratio of β-ion coincidences is unique to this ion-trap method and yields the highest precision result because many of the systematic uncertainties associated with the coincident detection cancel. This approach of studying βn emission allows measurement of both the energy spectrum and the branching ratio in the same experiment and is subject to different systematic effects than direct neutron detection experiments.

The results for ^{137}I and ^{144,145}Cs obtained here are consistent with previous measurements employing direct neutron detection. The results for ^{138}I confirm the more recent P_n measurements and put the βn decay data for this nucleus on much firmer footing. This should re-

duce concerns about using ^{138}I as a IAEA standard, as consistent βn-decay properties have now been established through the use of several independent measurement techniques.

Additional upgrades to the experimental setup can be implemented to increase the solid angle of the detector array and to improve the ion collection and cooling while minimizing the effect of the electric fields on the recoil ions. Larger β detectors and MCP ion detectors can increase the β-ion coincident detection efficiency by roughly a factor of four. Future experiments will also benefit from increases in the intensity and purity of the low-energy beams delivered by the CARIBU facility [49, 50]; there has been an order-of-magnitude increase in CARIBU beam intensities since this data was collected. Additional refinements to the electrode structure, including bringing the electrodes closer to the ion trap center, will allow lower-amplitude RF voltages to be used, which will reduce the perturbation of the ion trajectories prior to reaching the MCP detectors. These improvements will allow measurements to probe more neutron-rich isotopes, where the data are scarce, and the decay properties will provide valuable information on r-process nucleosynthesis and the structure of nuclei far from stability.

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